Content

Learning objectives and content

Objectives

At the end of the course, the student will be able to:

- describe the basic components (sensor, signal conditioner, controller, actuator) in a control system;
- describe the physical conversion phenomena underlying the operation of transducers (sensors, actuators);
- apply filtering and DSP techniques on the signals from and to the transducers;
- analyze the static and dynamic characteristics of transducers and the complete system;
- analyze the static errors that limit the accuracy of transducers;
- dimension the interface electronics used in a control system that contains sensors and actuators;
- select control algorithms that meet the performance requirements of the system under design;

Content

Almost any modern system contains sensors to observe the environment and actuators to influence this environment. An example of such a system is a modern car which is typically equipped with temperature, pressure, speed, chemical sensors and various mechanical values and engines. This course provides an introduction to sensor and actuator technology. It discusses the basic principles behind the most commonly used types of sensors and actuators. It also deals with the physical principles which underlie the operation of these sensors and actuators. The course provides also a basic introduction into the implementation of signal processing algorithms on embedded compute platform, control theory and interface electronics such that students can design a complete system which integrates the sensors and actuators introduced in this course.

2026

Sensing computing & actuating (5AIB0)

Practical information

Course module
5AIB0

Course type
Bachelor College

Credits (ECTS)
5 EC

Category
Advanced

Language of instruction
English

Offered by
Electrical Engineering - Electronic Systems

Course enrolment

OSIRIS Student
You can register yourself via OSIRIS Student. To do so, go to Enroll / Course.

Enrolment periods

- Block 4
  Timeslot(s)
  B - Mo 5-8, Th 9-10, We 1-4

  Enrolment period
  15 November 2026 until 21 March 2027 23:59

  Unenrolment
  until 19 April 2027 23:59

Entrance requirements

Assumed previous knowledge
5XCA0 Fundamentals of electronics 5EIA0 Computation1:hardware/software int.

Instructional modes

Instructional modes

- Lecture
  Remark

Tests

Tests

- Written examination
  Final examination

  Test weight
  100

  Minimum grade
  5.0

Test enrolment

OSIRIS Student
You can register yourself via OSIRIS Student. To do so, go to Enroll / Test.

Written examination

- Block 4
  Enrolment period
  15 November 2026 until 6 June 2027 23:59

  Unenrolment
  until 6 June 2027

  Test date
  2 July 2027

- Block I
  Enrolment period
  15 November 2026 until 1 August 2027 23:59

  Unenrolment
  until 1 August 2027

  Test date
  18 August 2027

Lecturer(s)

Contactperson for the course

- dr.ir. S. Stuijk

Responsible lecturer

- dr.ir. S. Stuijk

Subject matter expert

- dr.ir. S. Stuijk

Materials

Required material

- "Sensors and Actuators: Engineering System Instrumentation, Second Edition" by Clarence W. de Silva, CRC Press, ISBN 9781466506817

- All slides and exercises that are part of the course. This material is available as a download from the course website.

Do you want to know more?

dr.ir. S. Stuijk can tell you more about this course.
Send an e-mail to s.stuijk@tue.nl